



Ursinus
COLLEGE

CS375 Spring 2026

Ralph Rostock

Week 4

Requirements Validation, Stakeholders, Backlogs,
And Planning Realism



Ursinus
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Team Standup Meetings



When Software Projects Fail

- Mars Climate Orbiter
- Denver Airport Baggage System
- Healthcare.gov Launch
- Knight Capital Trading Loss

Did coding fail, or something earlier?



Requirements Validation = Risk Management

Validation Reduces:

- Technical Risk
- Schedule Risk
- Cost Risk
- Stakeholder Misalignment
- Rework



Professional Requirement Quality Framework

A High-Quality Requirement is:

- Atomic
- Unambiguous
- Verifiable
- Feasible
- Consistent
- Traceable
- Necessary
- Prioritized

Consider: “The system shall provide secure and fast login.”



Validation Is Collaborative

Stakeholder <----> Engineer

- Clarification
- Assumption testing
- Constraint discovery
- Trade-off discussion
- Negotiation



The Engineer's Question Toolkit

- What problem are we solving?
- Who is the primary user?
- What are edge cases?
- What constraints exist?
- What does success look like?
- What happens if we don't build this?



Refinement Changes Scope & Estimates

Before refinement:

“Prevent double booking” → 5 points

After refinement:

- Reduce rework cost
 - Align stakeholders
 - Prevent scope drift
 - Increase project confidence
- 13 points

(We’ll talk about “points” in a bit).



Stakeholder Role-Play Exercise

“Users must receive instant notifications via email, SMS, and push”

- Clarify
- Constrain
- Negotiate



Professional Negotiation Strategies

Engineers negotiate by

- Framing trade-offs clearly
- Using estimates and data
- Offering phased solutions
- Protecting constraints
- Avoiding adversarial tone

✗ “We can’t do that.”

✓ “If we include SMS, timeline extends by 2 weeks...”



Requirements -> Backlog

Requirements



User Stories



Product Backlog



Sprint Backlog



Story Points: What They Really Mean

Story Points Measure:

- Complexity
- Uncertainty
- Risk
- Relative Effort

NOT:

- Hours
- Days
- Individual Productivity



Why Not Linear Scaling?

If 1 = Trivial

And 10 = Extremely Complex,

Is 8 really just “a little more” than 7?



Why Fibonacci Scaling Works

1 – 2 – 3 – 5 – 8 – 13 – 21

Each step increases uncertainty tolerance.
As complexity increases, precision decreases.



What Happens at 13 and Above?

If a story is 13 or higher

- Too large
- Too uncertain
- Poorly understood
- Needs to be broken down



Example: Reservation System Stories

Consider these 4 user stories, and assign story points

- Display available rooms 2
- Create reservation 5
- Prevent double booking with conflict detection 8
- Add recurring reservations 13



Velocity and Forecasting

Velocity = 12 points per sprint

Backlog = 44 points

Forecast $\approx$ 3.7 sprints

Forecast $\neq$ Guarantee



From Backlog to Timeline Validation

Planning Steps

- Break stories into tasks
- Estimate duration
- Identify dependencies
- Build timeline
- Identify critical path



What the Gantt Reveals

- Dependencies create bottlenecks
- Critical path determines finish date
- Total duration $\approx$ 22-24 days
- Available time = 20 days

Points says "Yes" 

Timeline says "No" 



What is the Critical Path?

The Critical Path

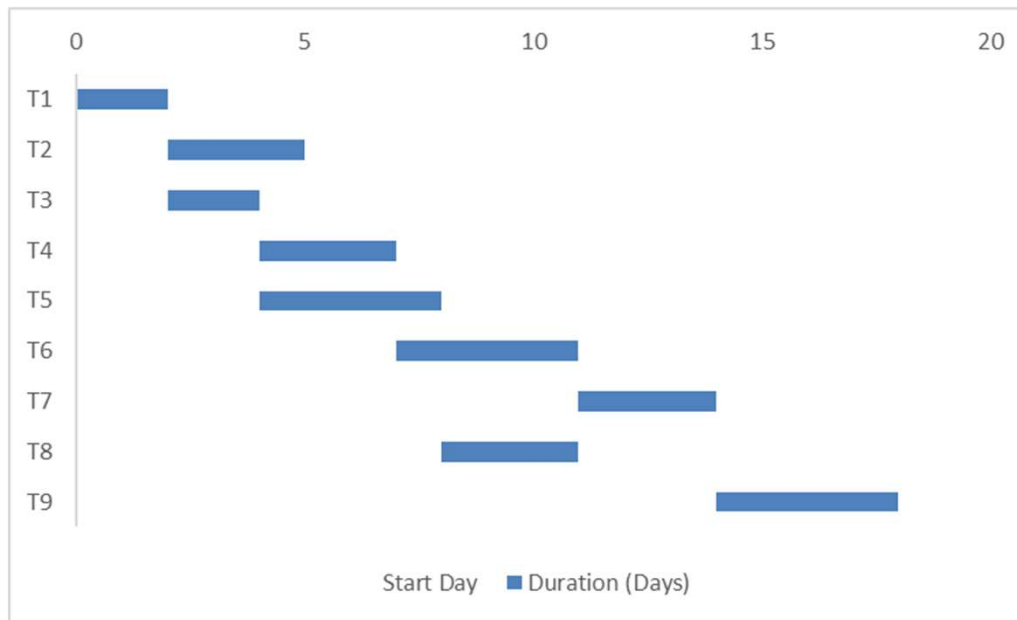
- The longest chain of dependent tasks.
- Determines the minimum possible project duration.

Key Properties

- If any task on the critical path is delayed, the project is delayed
- Tasks not in the critical path have **slack**.



Finding the Critical Path



Step 1:

Identify Final Task (T9)

Step 2:

Trace Dependencies Backward

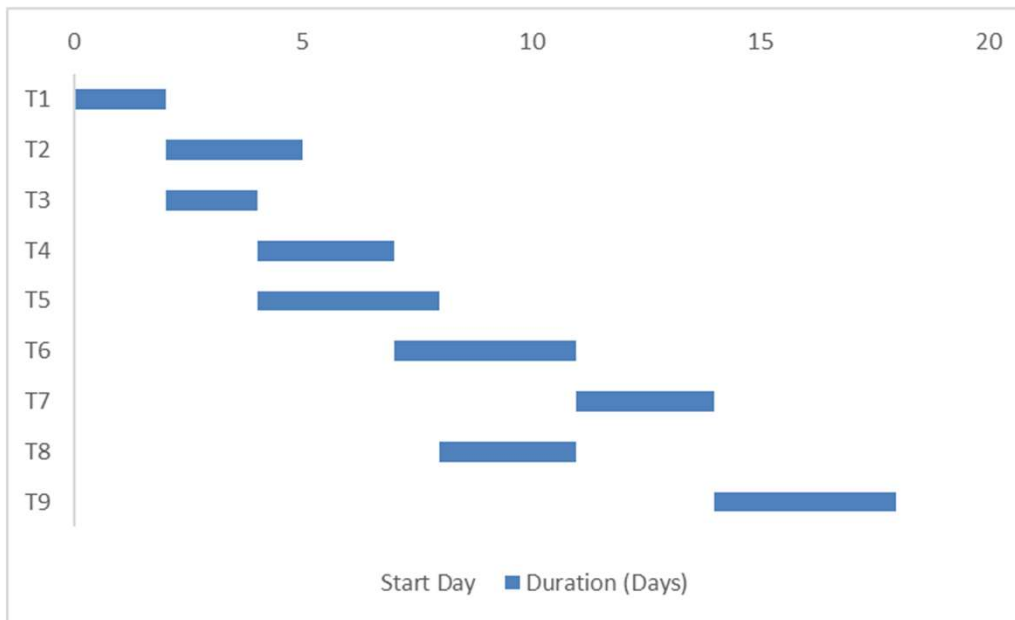
T9 ← T7 ← T6 ← T4 ← T3 ← T1

Critical Path:

T1 → T3 → T4 → T6 → T7 → T9
= 18 days



What is NOT Critical?



Not on the critical path:

- T2 (UI Design)
- T5 (Event Creation)
- T8 (Admin Dashboard)

These tasks have slack.



Why This Matters

Critical Path = Schedule Risk

It helps you:

- Focus monitoring
- Prioritize resources
- Negotiate scope intelligently



Data-Driven Negotiation

Engineer says:

“If we include advanced conflict detection and email integration, the timeline exceeds our deadline.”

Options:

- Remove email from MVP
- Simplify conflict detection
- Phase admin features
- Extend timeline



Backlog + Gantt: Complementary Tools

Backlog answers “What should we build?”

Gantt answers “Can we build it in time?”

Validation requires both.



Final Takeaways

A requirement is not complete until

- It is validated
- It is refined with stakeholders
- It is prioritized
- It is estimated
- It fits the schedule

Engineering = disciplined decision-making under constraints



Assignment: Weekly Standup Reflection

<https://rrostock1.github.io/Ursinus-CS375/Assignments/StandupReflection>

(Individual Assignment)

Due Feb 19 (and every week thereafter)



Assignment: The Overdose

<https://rrostock1.github.io/Ursinus-CS375/Assignments/Overdose>

(Individual Assignment)

Due Apr 23



Assignment: Requirements Report Document

<https://rrostock1.github.io/Ursinus-CS375/Project/Requirements>

Due Feb 26



Due Next Week (Feb 26, by Midnight)

- Weekly Standup Reflection
- Software Requirements Document
- Git Homework Assignment